Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 10: missing information cannot automatically satisfy a condition.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Detect absent and unusable values. Try the recall before opening notes.

Your learning target

I can detect absent and unusable values.

I can keep unknown distinct from zero.

I can report accepted and rejected records.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

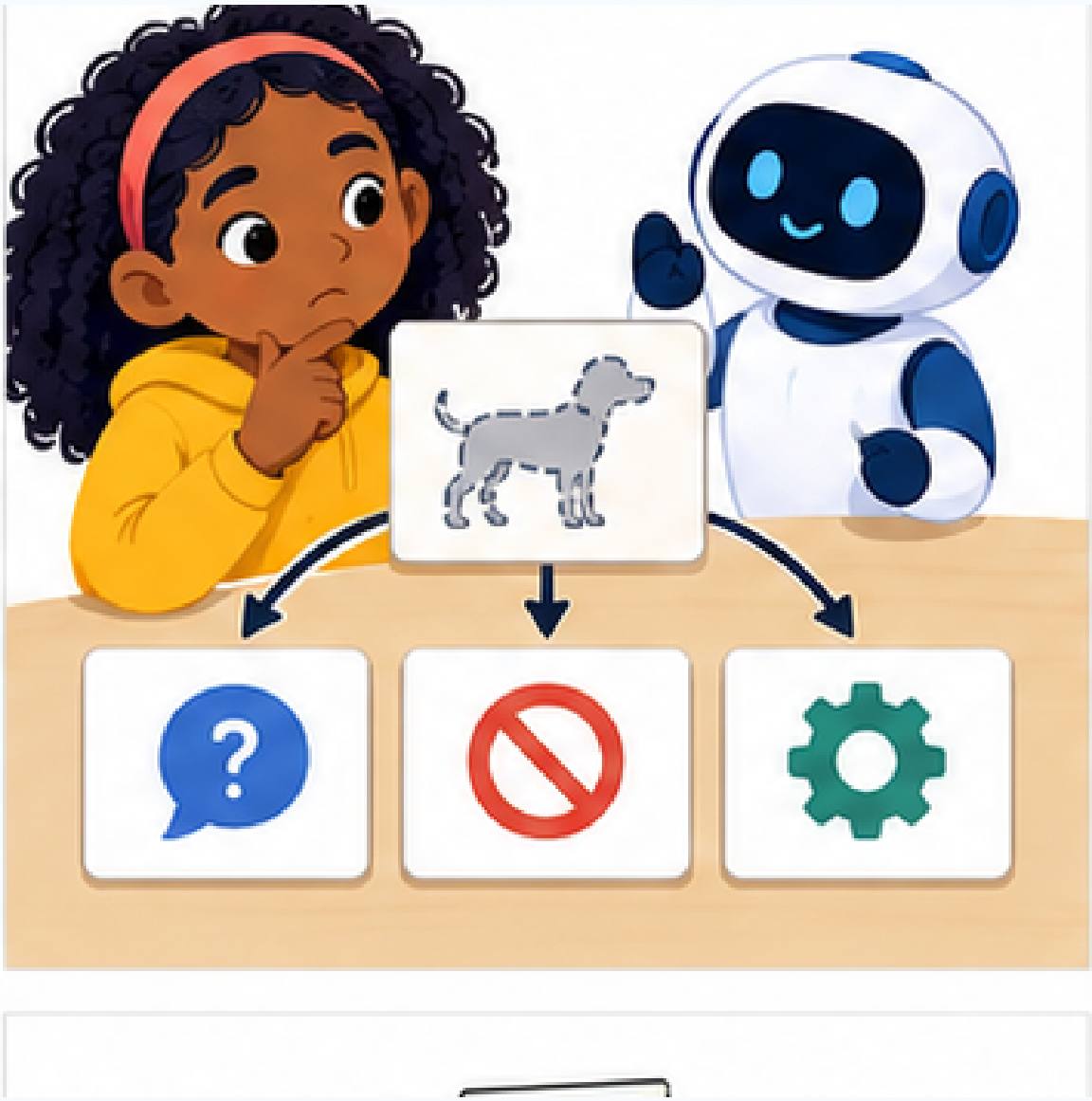
Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 18: **Missing Data**

A system needs an explicit policy for unknown fields.



THINKER MISSION

Compare ask, **abstain**, and **default** strategies.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Is the text "12" the same type as the integer 12?

This week's end goal

By the end, I can validate **missing value**s without confusing zero with absent data.

STUDENT LESSON · 2 OF 18

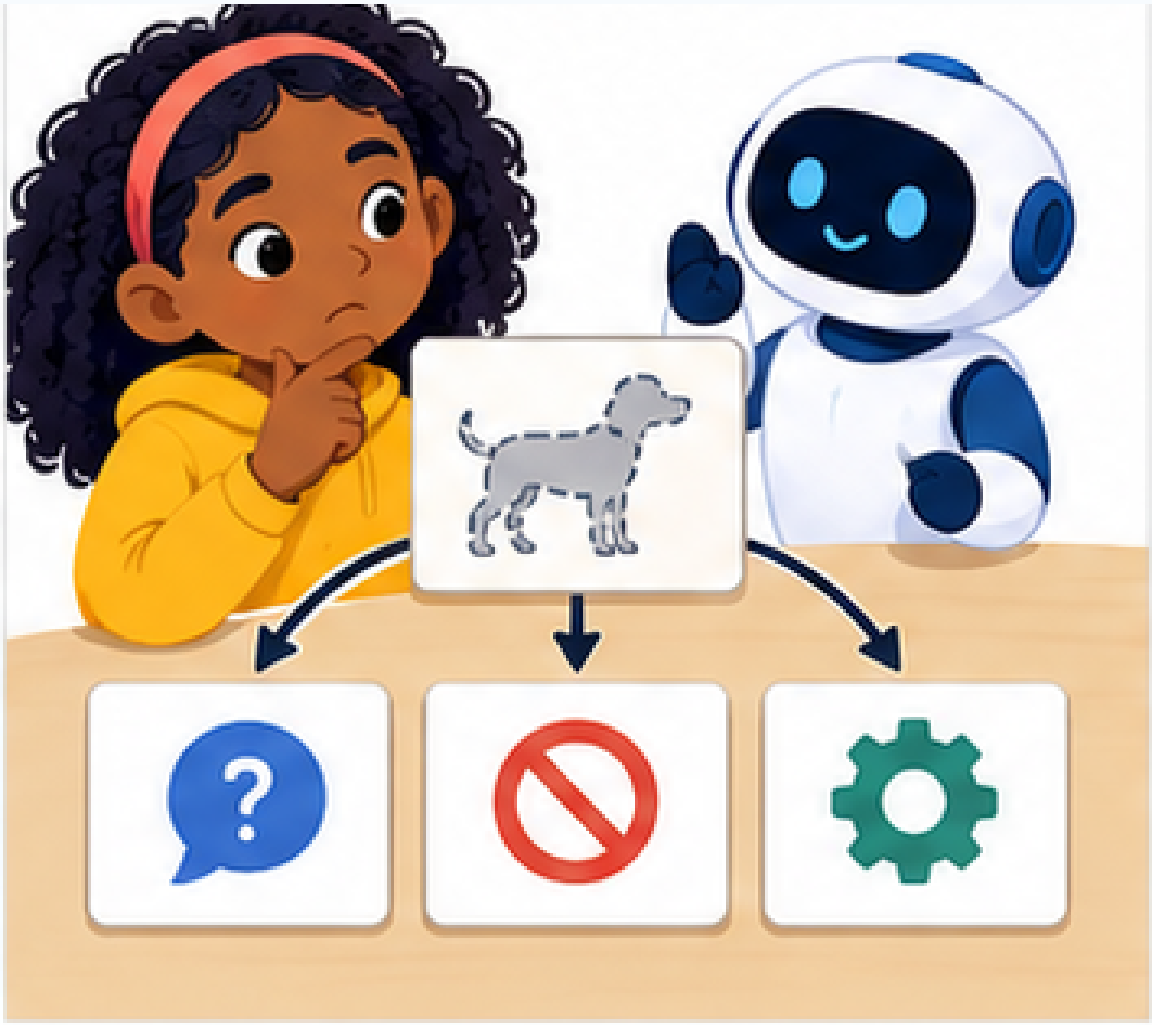
Notice the evidence

> Investigate the supplied Missing Data case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **missing data** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses.

STUDENT LESSON · 3 OF 18

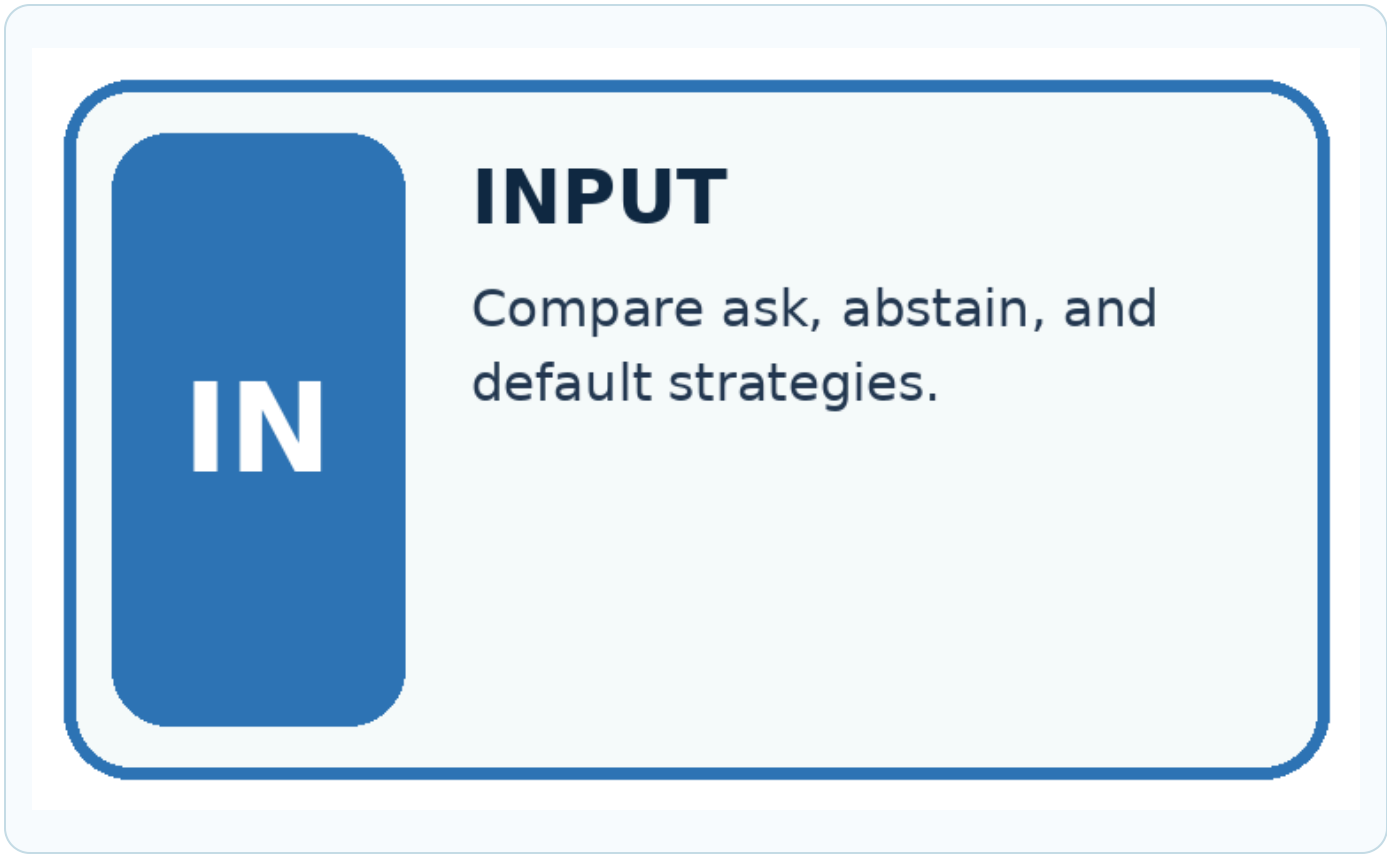
See how it works

> Follow the mechanism for Missing Data. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

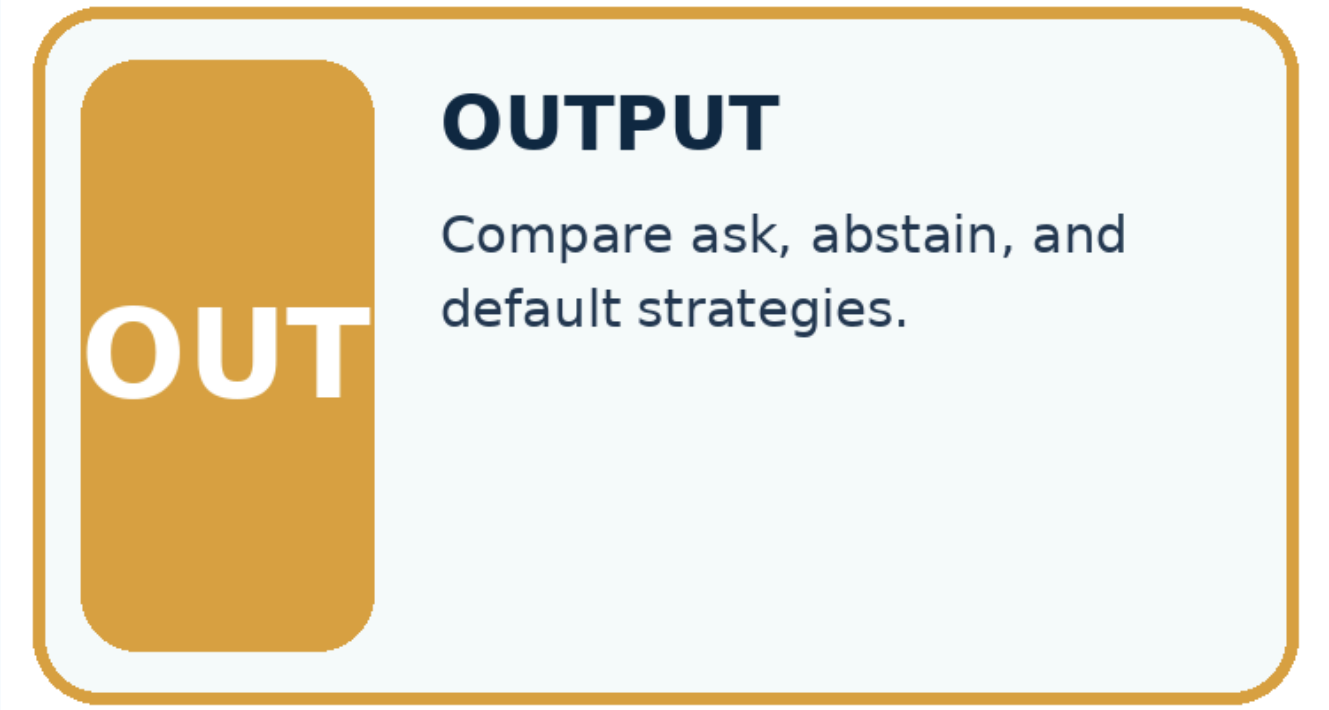
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare ask, abstain, and default strategies.

↓


PROCESS
A system needs an explicit policy for unknown fields.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
rows = [{"minutes": 0}, {}, {"minutes": 12},
        {"minutes": -1}, {"minutes": None}]
accepted = []
rejected = 0
for row in rows:
    value = row.get("minutes")
    if type(value) is int and 0 <= value <= 60:
        accepted.append(value)
```

```
    else:
        rejected = rejected + 1
print(accepted)
print(rejected)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Recorded zero	0	Accept as a measurement
Absent or None	Unknown	Hold for review
Negative integer	Outside range	Reject under this schema

Read each row from left to right and connect the evidence to the stated rule.

dict.get returns None when this key is absent. An explicit None also represents unknown in this dataset. append adds an item to the end of a list. The script prints [0, 12] and then 3.

The exact type check prevents comparing None or text with the numeric range.

Keep zero separate from unknown

A recorded 0 says the measured duration was zero. A missing field says we do not have the duration. They are different statements. Trace each record with three columns: raw value, decision and reason.

Keep 0 and 12; reject the absent value, -1 and None. The accepted list therefore has two values and the **rejection** count is three. Do not turn an unknown value into a number just to make the calculation run.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

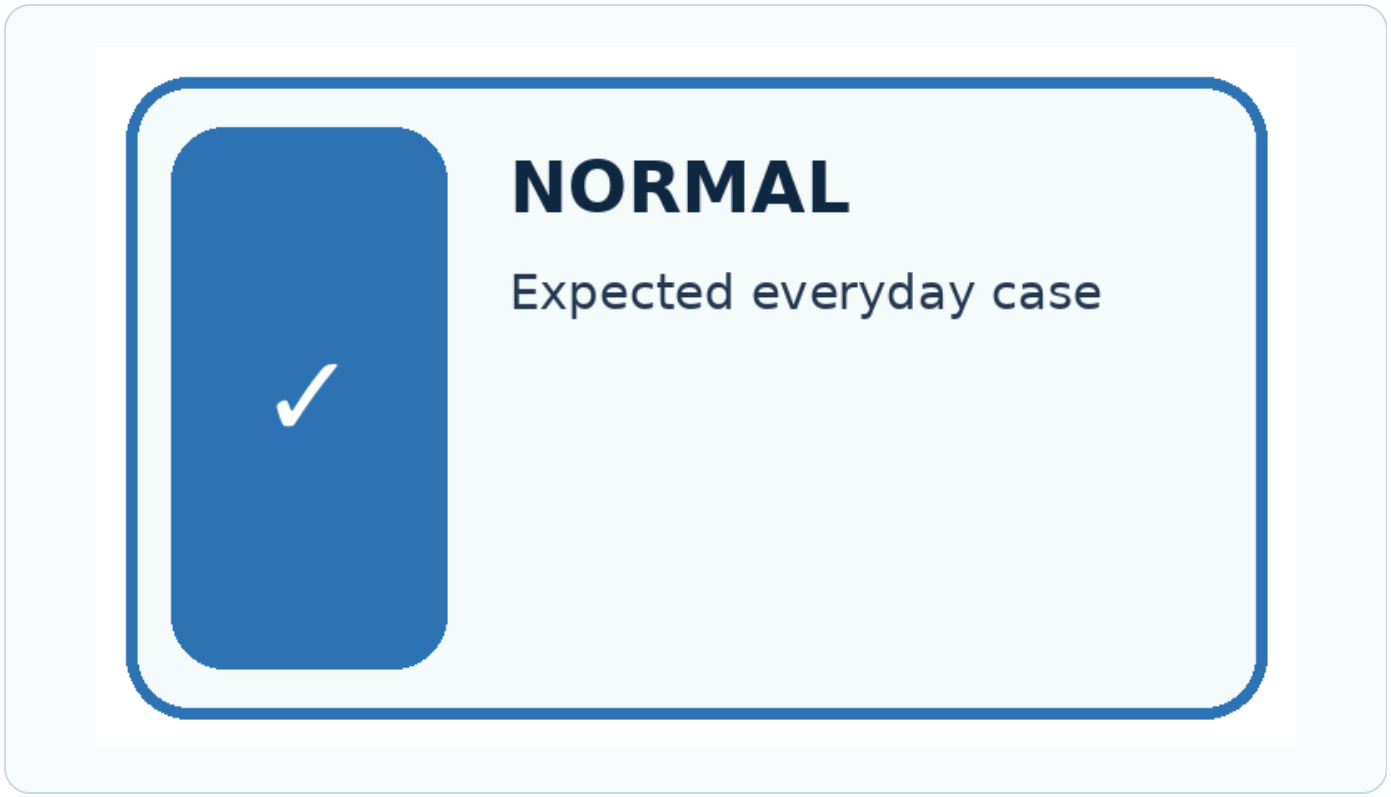
> Compare the supplied cases for Missing Data. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

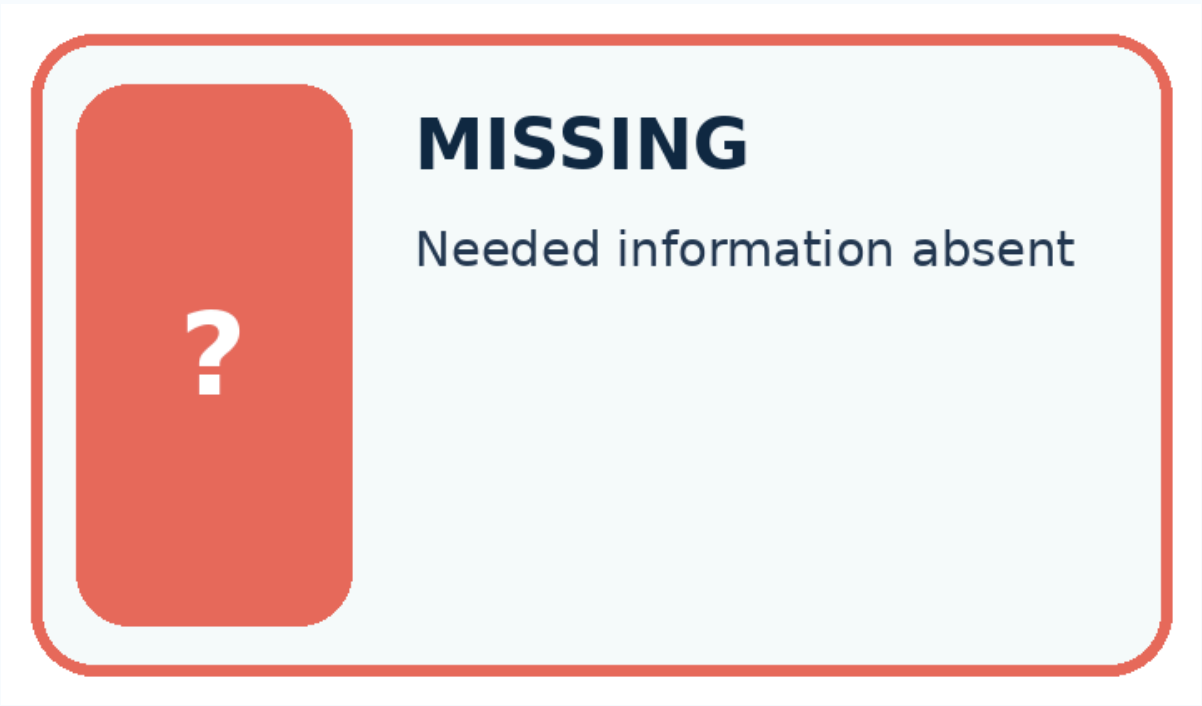
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Replace text "4" with integer 4 through an authorised conversion.

Predicted result

Prediction: Accepted values become [0,4,4]; three rows remain rejected.

Why the result follows

Reason: The converted value now satisfies both type and range checks.

STUDENT LESSON · 5 OF 18

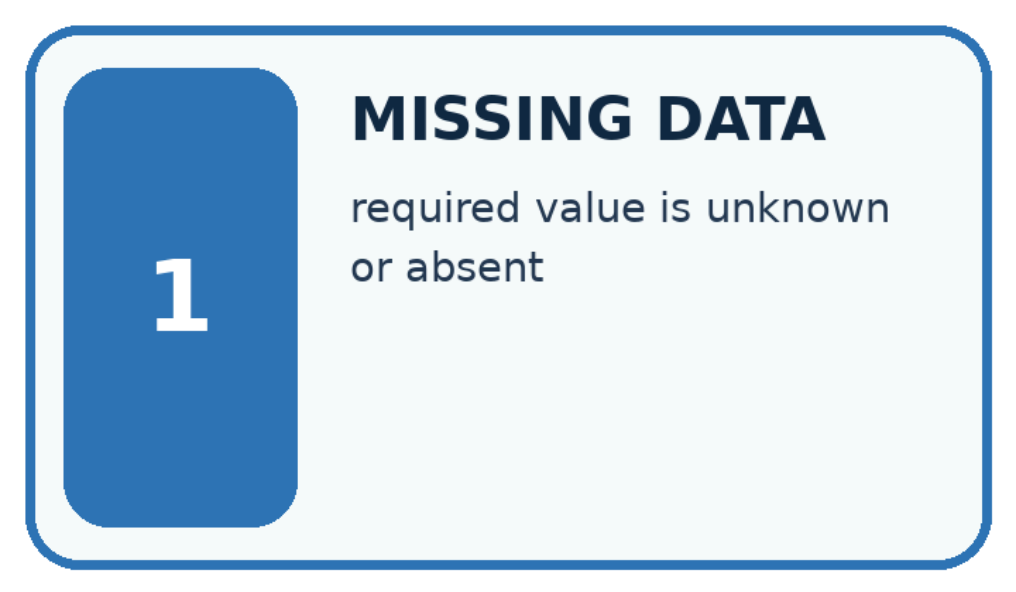
Words that unlock the idea

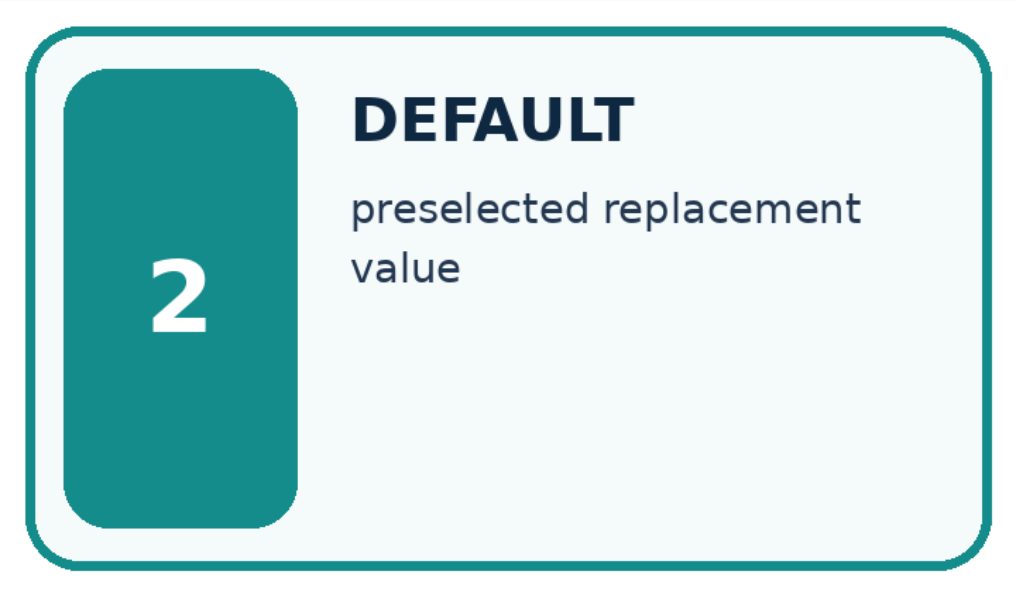
> Use the vocabulary of Missing Data to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

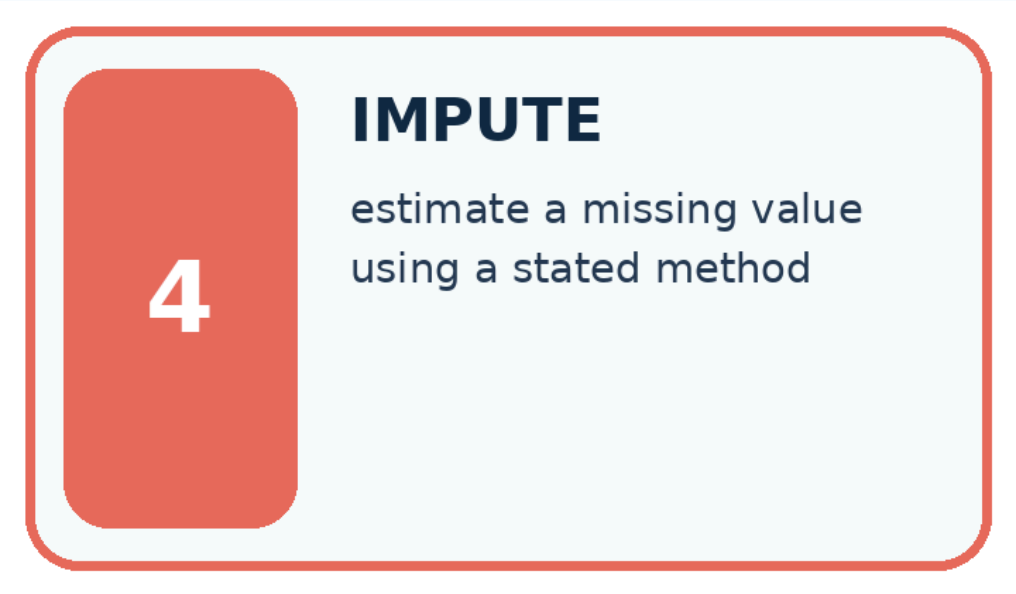
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
MISSING DATA
required value is unknown or absent


2
DEFAULT
preselected replacement value





Word	My own example	Not this / boundary
missing data	Information that was not recorded or is unavailable, which is different from an observed zero.	
default	The value or behaviour used when no alternative is specified.	

Word	My own example	Not this / boundary
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.	
impute	Replace a missing value using a stated method; an imputed value is an estimate, not an observation.	

Words used in the closer explanation

Word	Meaning
Missing value	A value that was not recorded
Validation	Checking data against explicit rules
Rejection	Holding a record out because it fails a rule
Coverage	Which intended records or groups are represented

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Missing Data, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 18 • WORKED EVIDENCE

1. Reader’s length preference is missing.
2. Ask: best when the user can answer.
3. Abstain: safest if the field is essential; default: convenient but may misrepresent.
4. Record which policy was used so the explanation stays truthful.

1. Reader’s length preference is missing.

2. Ask: best when the user can answer.

3. **Abstain**: safest if the field is essential; **default**: convenient but may misrepresent.

4. Record which policy was used so the explanation stays truthful.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Add {"minutes": 60} and {"minutes": "5"}. Predict the accepted list and rejection count.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Missing Data

A fictional catalogue records counts of available copies under a strict **input** contract.

The facts we will use

Supplied facts

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Supplied facts

Rows have count 0, count 4, absent count, count "4", count True and count 9.

AI ACADEMY / AI-2/18

Missing Data

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Retrieve carefully
Use row.get("count"); absent field gives None, distinct from 0.

↓

2. Check exact type
Integers 0,4 pass; text "4", Boolean True and None fail.

↓

3. Check range
Integer 9 fails maximum 8.

↓

4. Report validation
Accepted counts [0,4]; four rows rejected with recorded reasons.

CHANGE ONE THING
Replace text "4" with integer 4 through an authorised conversion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve carefully	Use row.get("count"); absent field gives None, distinct from 0.
2. Check exact type	Integers 0,4 pass; text "4", Boolean True and None fail.
3. Check range	Integer 9 fails maximum 8.
4. Report validation	Accepted counts [0,4]; four rows rejected with recorded reasons.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Testing if value would wrongly discard zero, which is a real valid count. Python’s isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

What this example does not establish

Keep the stated limit

Type and range checks cannot confirm whether the real shelf count was measured accurately.

Keep the stated limit

Read the labelled picture: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Keep the stated limit

Change one thing: Replace text "4" with integer 4 through an authorised conversion. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Missing Data claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Blank values can safely be treated as zero without explanation.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Rejection can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

> **Only 2 of 5 supplied durations were accepted. Why should the report include this?**

STUDENT LESSON · 8 OF 18

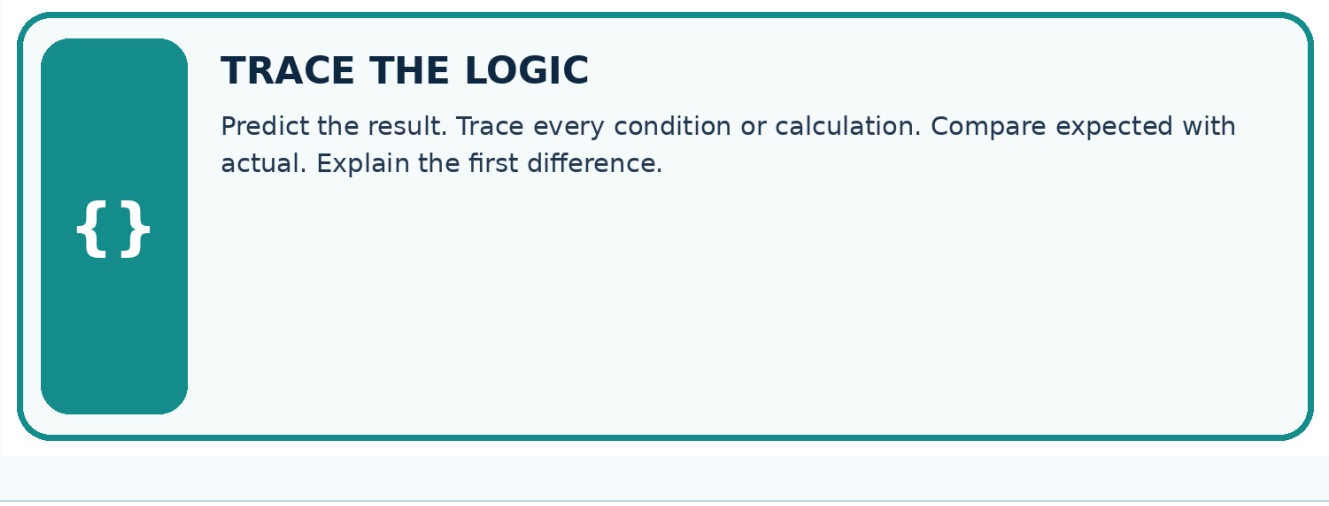
Practise together

> Practise Missing Data with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Using if value would discard the legitimate zero and accept some invalid nonempty text. Test the written type and range rules explicitly. Filling an absent duration with 0 falsely claims an observed zero.

Practise with less help: Trace filtering

Use workbook W2 for the connected example "**Missing value**s and **validation**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Missing value**s and **validation**. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

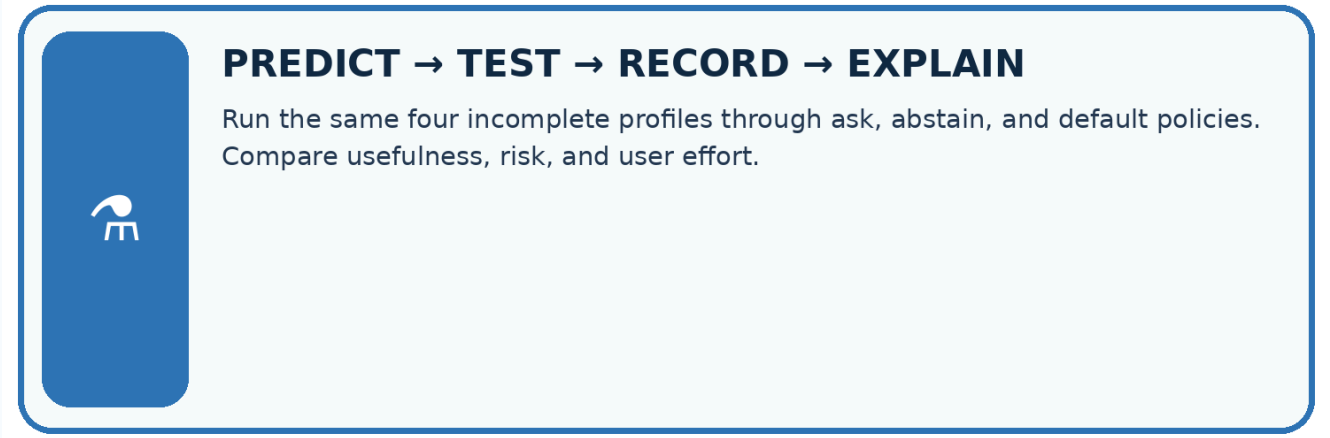
Run the investigation

> Investigate Missing Data. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Run the same four incomplete profiles through ask, **abstain**, and **default** policies. Compare usefulness, risk, and user effort.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

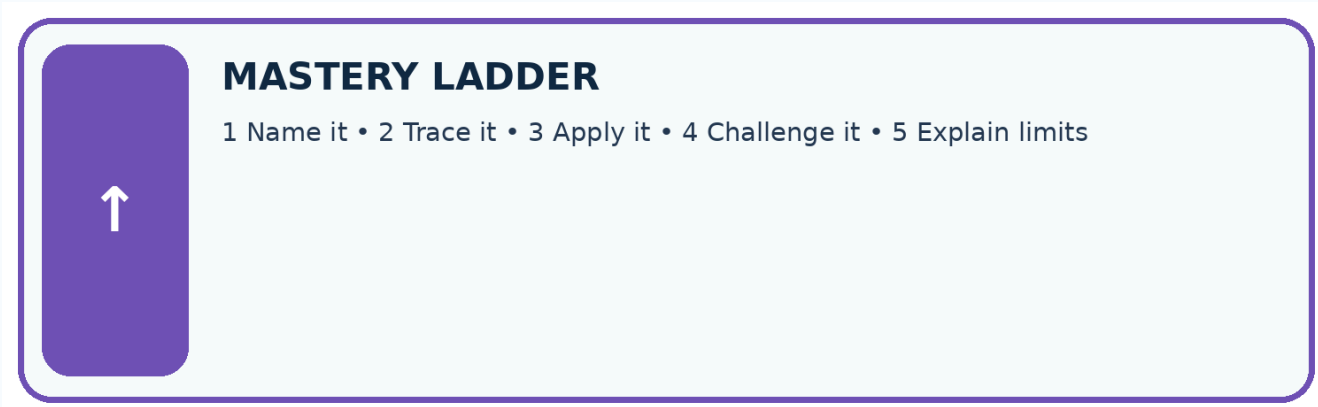
Build your understanding

Move from recall to tracing, diagnosis and transfer in Missing Data. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

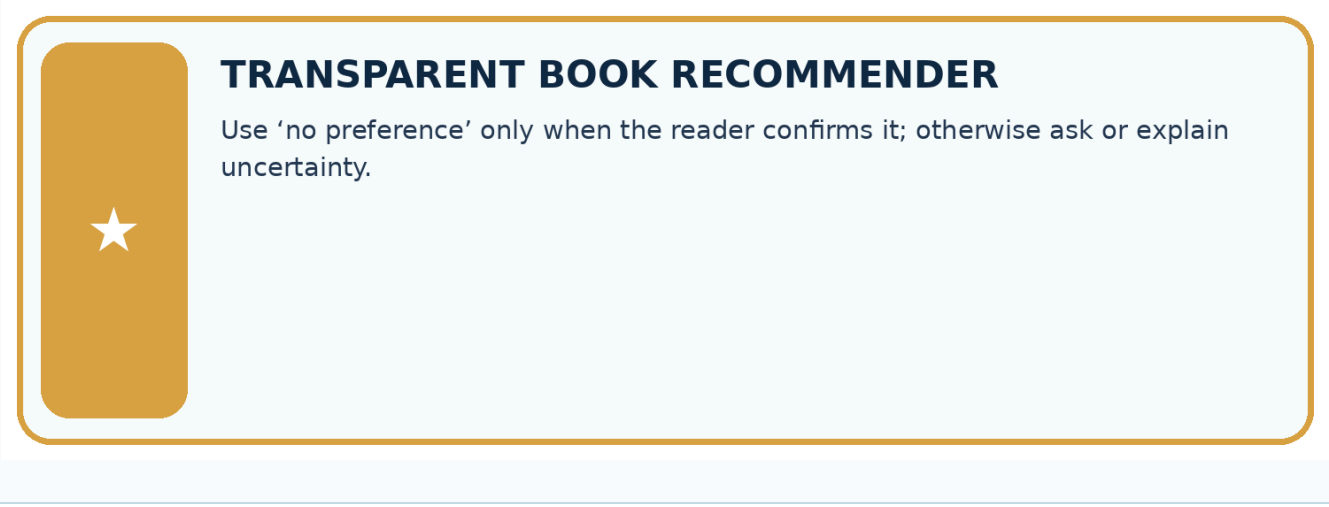
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

STUDENT LESSON · 12 OF 18

Show what you know

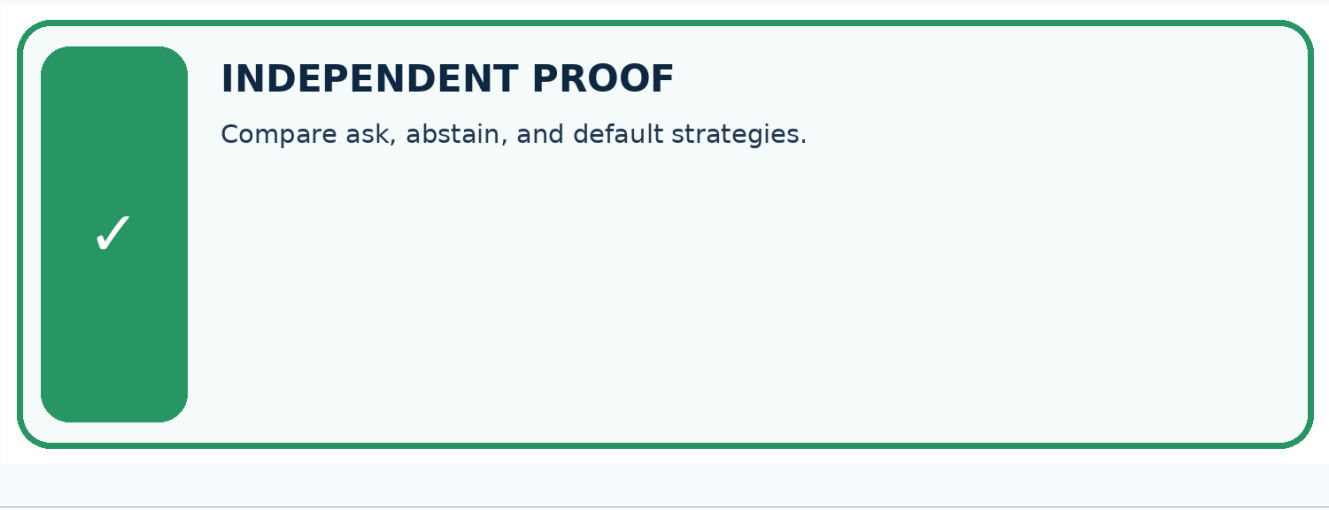
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compare ask, **abstain**, and **default** strategies.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Missing Data case for supported practice. Repeated examples do not become fresh assessments.

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Genre is known, but page count is missing and length affects recommendation score.

The question

Choose transparent handling.

Worked explanation

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Missing is different from zero

Evidence or stage	Value or result
Retrieve	Look up the page count
Ask	Request the missing information

Evidence or stage	Value or result
Partial score	Use a documented policy with uncertainty

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Validate independently

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Missing Data, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional catalogue records counts of available copies under a strict **input** contract.

By the end, I can	How I will show it
Detect absent and unusable values	Explain the deciding step, show a trace or test, and state one limit.
Keep unknown distinct from zero	Explain the deciding step, show a trace or test, and state one limit.
Report accepted and rejected records	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Missing Data without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Detect absent and unusable values

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue records counts of available copies under a strict **input** contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accepted counts [0,4]; four rows rejected with recorded reasons.

1. Report **validation**

2. Retrieve carefully

3. Check range

4. Check exact type

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report **validation** Accepted counts [0,4]; four rows rejected with recorded reasons. Testing if value would wrongly discard zero, which is a real valid count.

Python’s isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace text "4" with integer 4 through an authorised conversion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Accepted values become [0,4,4]; three rows remain rejected.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Repair a missing building block

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

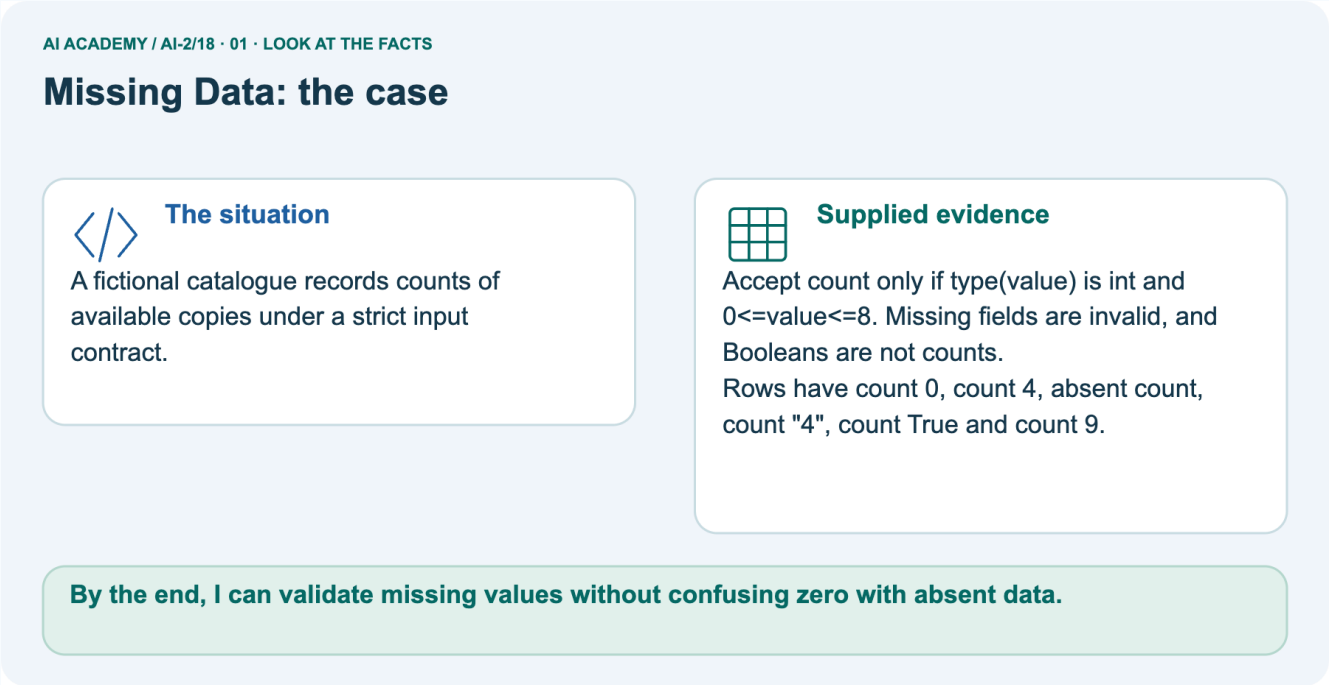
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Missing Data graphic using its labels, deciding evidence and relevant limitation.

01. Missing Data: the case



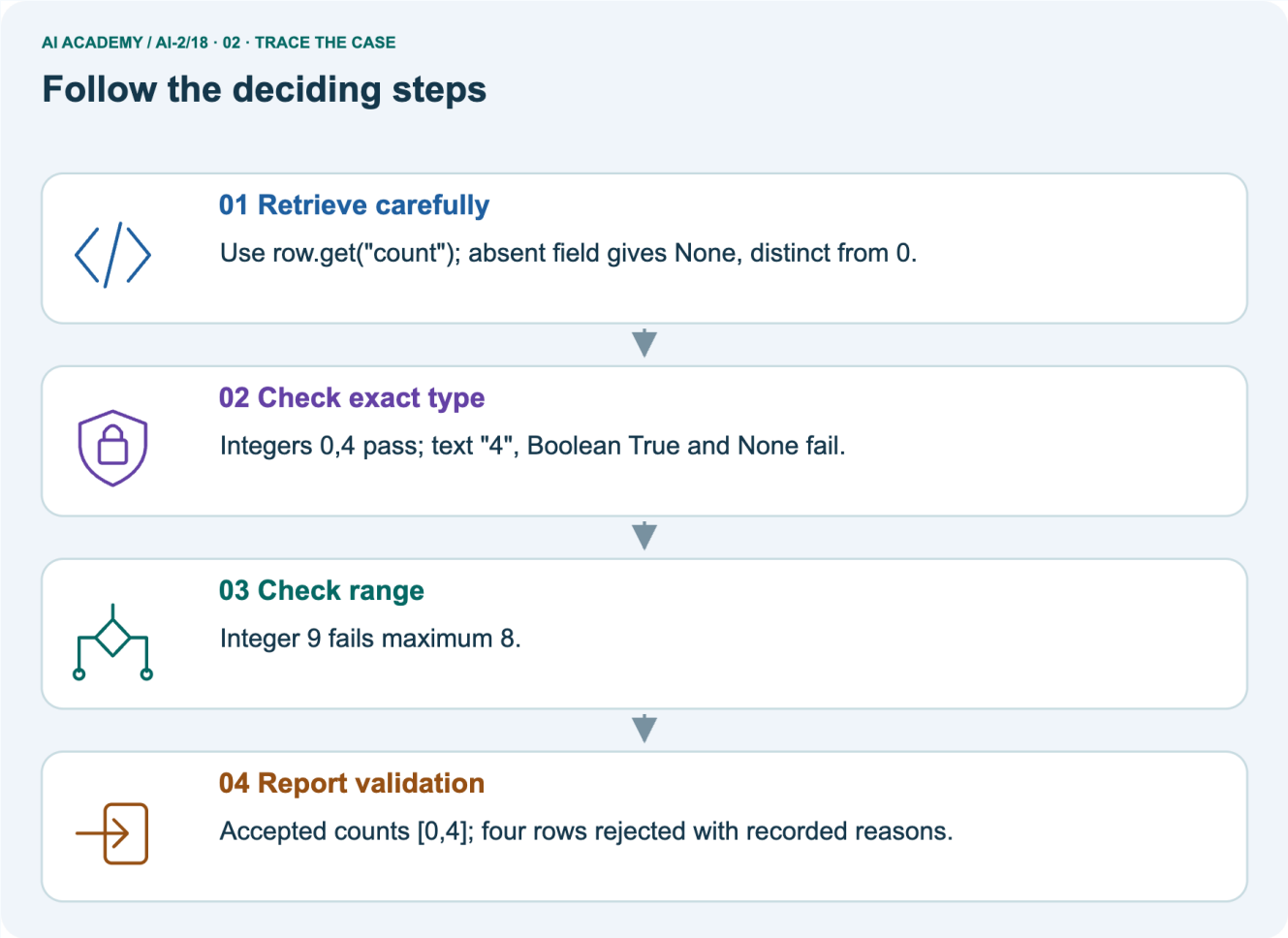
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue records counts of available copies under a strict **input** contract. Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

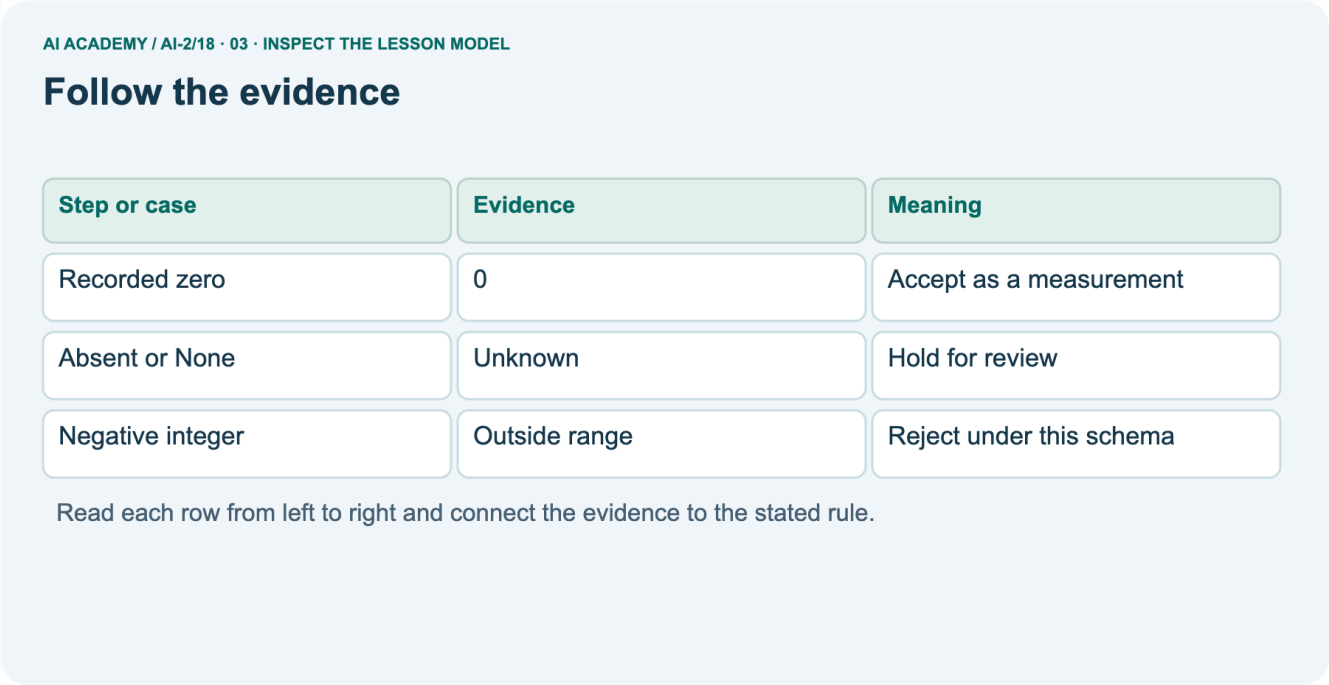


Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

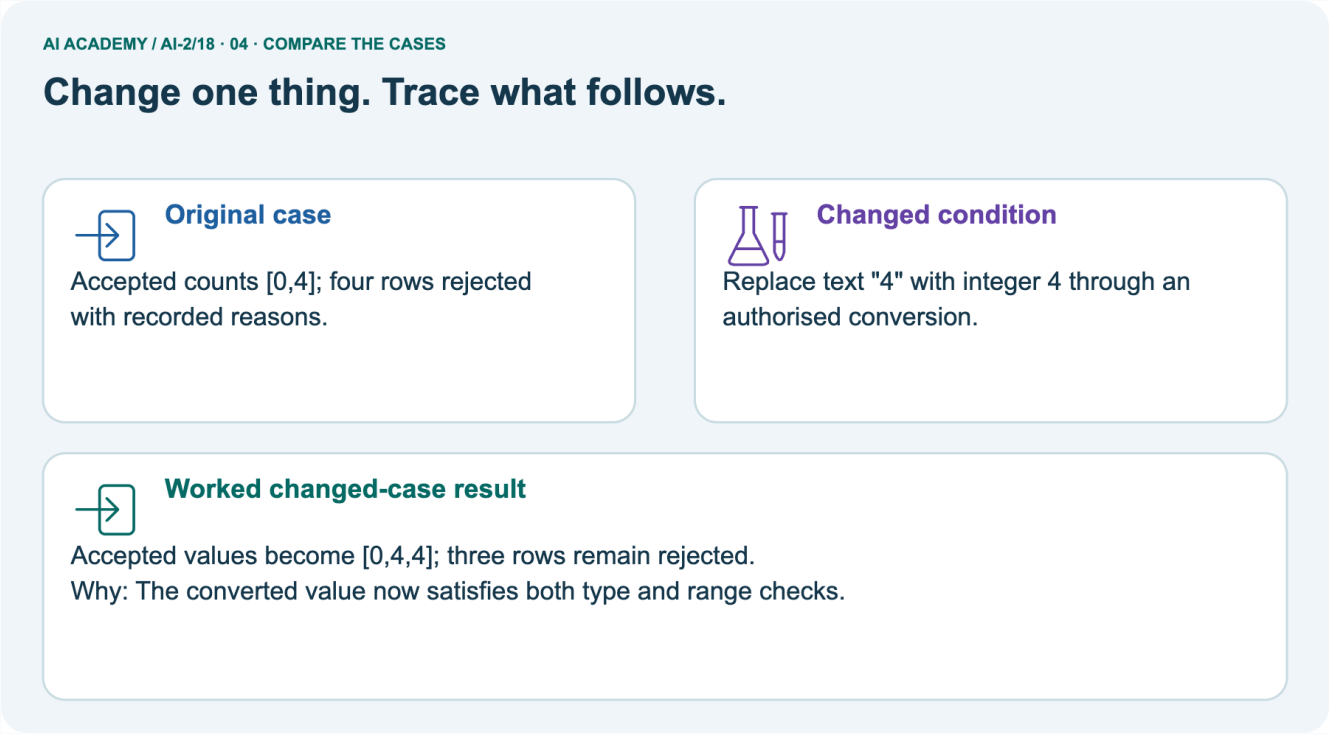


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



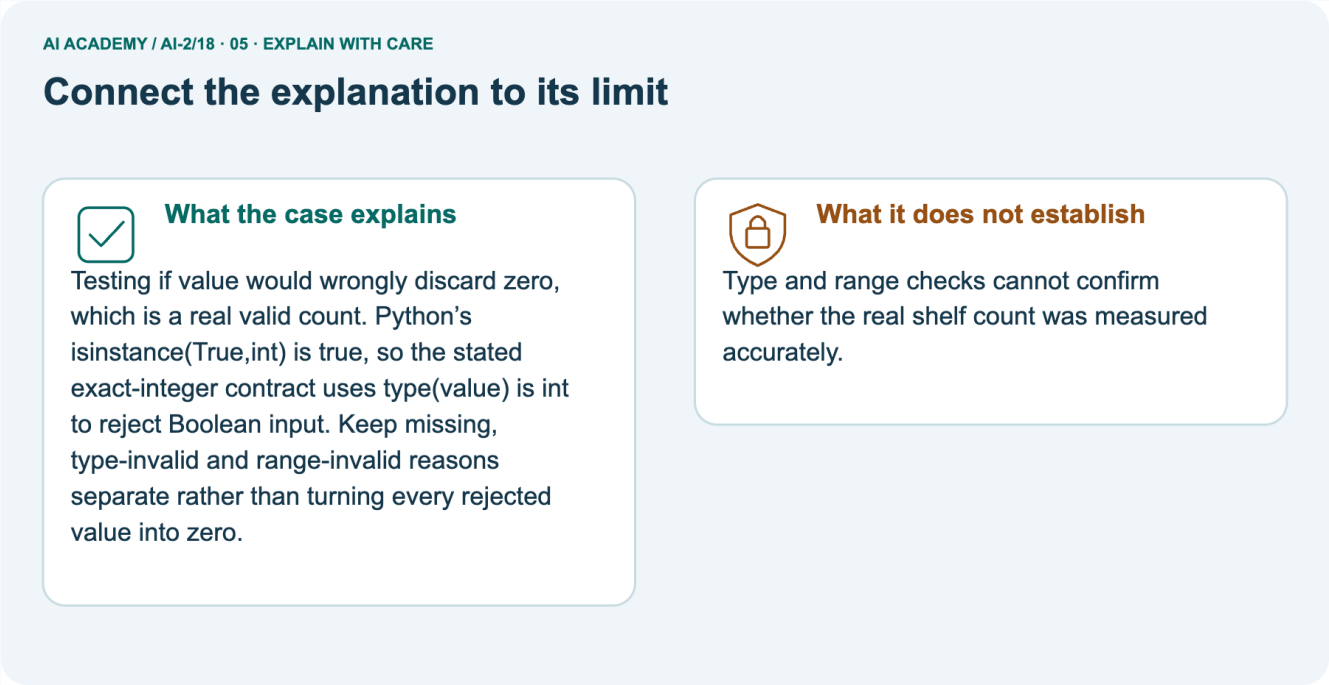
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



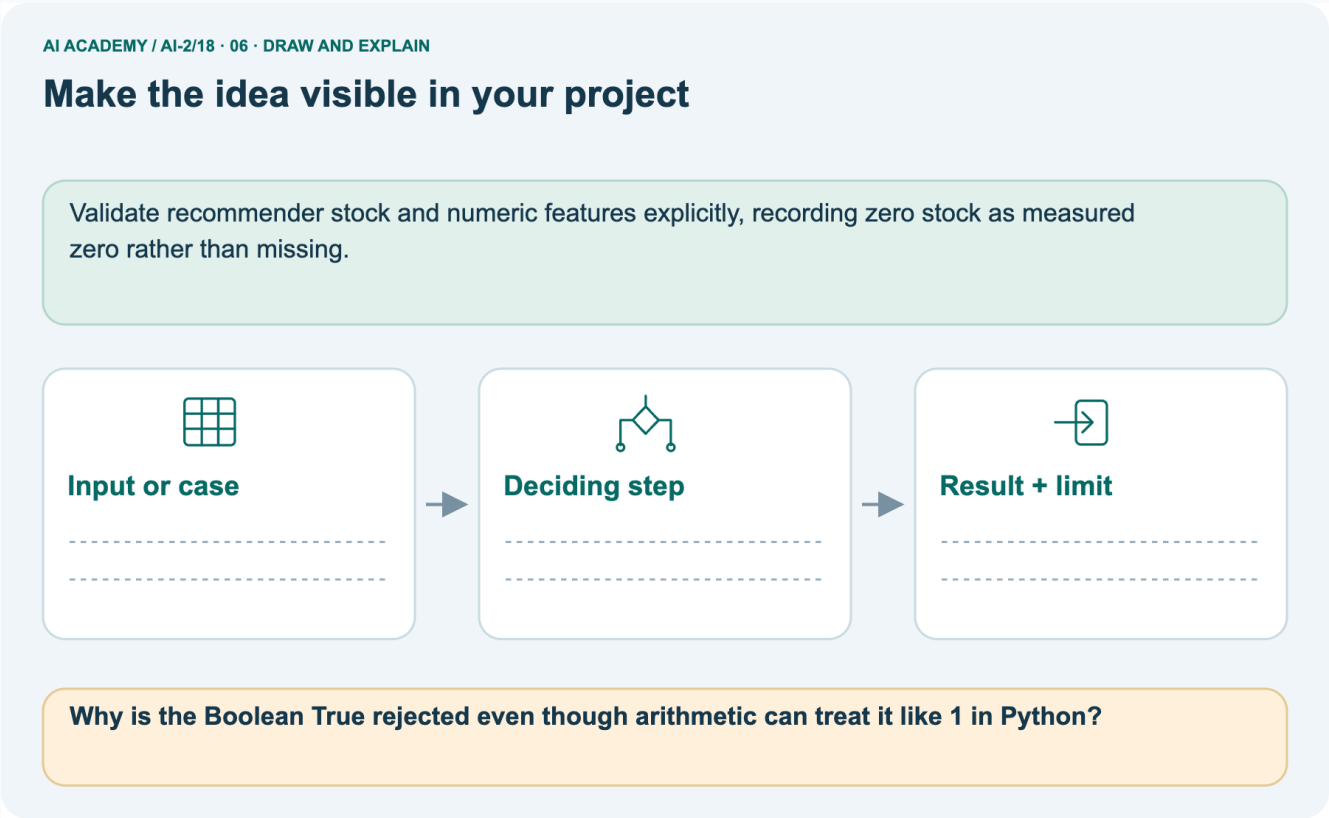
Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.